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Computed Tomography of Chemiluminescence (CTC): High resolution and instantaneous 3-D measurements of a Matrix burner

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Abstract

Computed Tomography of Chemiluminescence (CTC) is a diagnostic technique that provides instantaneous 3-D information on flame geometry and excited species concentrations. The technique reconstructs the 3-D chemiluminescence intensity field using Computed Tomography (CT) from integral measurements of chemiluminescence in the form of camera images. The CTC sensor has been demonstrated for a methane–oxygen Matrix burner consisting of 21 laminar diffusion jet flames using affordable *commodity* cameras. High resolution 3-D reconstructions obtained from 48 views show good agreement with the observed flame shape and resolve wavelengths of approximately 220 μm. This is sufficient to capture the multiple flame fronts, showing the suitability of CTC for wrinkled turbulent flames. Instantaneous 3-D reconstructions using 10 simultaneous camera measurements also show good agreement with the observed flame shape and are seen to be tolerant of error in the camera location. Measurements with exposure times as short as 62 μs were found to achieve more than sufficient signal-to-noise ratios for successful tomographic reconstructions. Obvious applications for CTC are the measurement of (instantaneous) flame-surface density, wrinkling factor, flame normal direction, and possibly heat release, as well as the observation of transient developments.
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1. Introduction

The development of better combustion devices requires ever more detailed and comprehensive experimental investigations of complex, transient (turbulent) flames. The availability of sensitive high-speed CMOS/CCD cameras and powerful lasers has enabled line and planar measurements of turbulent flames to be obtained with high temporal resolutions. However, turbulent flames are inherently 3-D and transient, so that time-resolved 3-D data are needed. Such data can be measured by (fast) scanning of planes [1,2], but this requires expensive high-speed cameras, while the requirement to position the detector normal to the plane of interest can be hard to satisfy. Computed Tomography (CT) techniques are robust, allow flexibility in detector locations (in 3-D), and can be implemented with affordable cameras, as will be demonstrated.

The Computed Tomography of Chemiluminescence (CTC) is a non-intrusive sensing technique that provides the 3-D locally resolved chemiluminescence intensity of a flame. The technique uses a CT algorithm to recover the intensity

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field from projection measurements taken simultaneously by multiple cameras from different viewing angles. The more images are taken, the higher the resolution in the recovered field will be. Through the use of commodity hardware rather than low-volume scientific cameras, there is the potential for the deployment of large numbers of cameras and hence high resolutions.

In combustion applications CT has previously been applied to refractive index measurements [3–7] and laser absorption measurements [8–14]—albeit only in a single plane due to the difficulty of providing multiple laser sources and detectors. In contrast, because CTC measures natural emissions, external sources are not required, thereby allowing simpler, more robust implementations, with no need to ensure the precise source-detector alignment. The coupling of CT with natural emissions has been considered previously for *in-situ* control purposes [15–18], and to measure single laminar flames [19–21]. Ishino and Ohiwa [22] used a custom-built 40 lens film camera to produce high resolution 3-D reconstructions of the visible emissions of a flickering laminar flame.

Chemiluminescence occurs when light is emitted as a result of chemical reaction. Because the wavelength emitted depends on the specific transition that occurs [23], chemiluminescence also provides a measure of the concentration of that particular excited state species. This work considers visible chemiluminescence which stems mainly from the CH^* ($A^2\Delta - X^2\Pi$) transition and which has negligible self-absorption [19,24].

Chemiluminescence measurements have been used to study local equivalence ratios [25–29] and heat release [26,29], as well as transient phenomena like flash-back [2] and auto-ignition [30]. Recently, much interest has been shown in the modelling of chemiluminescence [31,32,28,29] and such work may yield further relationships between chemiluminescence and relevant combustion parameters. As chemiluminescence occurs near the narrow *reaction zone*, CTC also provides information on the 3-D flame geometry, which is important for advanced modelling approaches like LES with flamelet or flame-surface density techniques that permit realistic flame visualisations like those presented by Stein et al. [33].

This paper reports two sets of 3-D chemiluminescence measurements that demonstrate the applicability of CTC to turbulent flames. Measurements of high spatial resolution are presented along with instantaneous measurements obtained from affordable machine vision cameras.

2. CTC experimental method

The CTC system comprises the camera arrangement and the CT software algorithm which reconstructs the intensity field. In the general CTC arrangement, the burner is located at the centre of a circle of N_q detectors, which each provide a 2D set of N_p projection measurements of the flame from a common viewing angle θ. Each set of projections is termed a *view*, and greater numbers of views allow higher resolution reconstructions.

2.1. CT processing

The CT technique can reconstruct a discrete estimate of a scalar field from any set of projection measurements, taken at different viewing angles, provided that the projection geometries are known.

This work relies on the iterative ART (Algebraic Reconstruction Technique) algorithm suggested by Gordon et al. [34], which has been used in many studies [19,35,36]. More advanced CT algorithms exist that may be better for specific setups. However, ART offers a combination of advantages that make it a particularly good choice for our application: algorithms of the ART family are robust in the presence of noise and can reconstruct a field from few views (optical access is often limited); ART can be adapted to incorporate *a priori* information (e.g., smoothness or boundedness) to improve reconstructions; and ART places no restrictions on the geometry of the projections nor on their relative orientation. Indeed, the geometric flexibility of ART is not found in most other algorithms, and permits its efficient extension into three dimensions, which in turn allows important 3-D optical effects to be considered, such as depth-of-field and beam divergence.

The ART algorithms are based on the expression of the reconstruction problem as an under-determined system of linear equations. The system is obtained by approximating each projection measurement I_{qp} as a finite sum taken through the discrete 3-D object f (i.e., the flame) domain as shown in Eq. (1). In this work the domain is divided into voxels, although several studies, e.g., [37,38] use other basis functions to exploit *a priori* information. To simplify the notation, the 3-D object indices are vectorised and represented by a single index v of maximum value N_v.

$$I_{qp} = \sum_{v=1}^{N_v} w_{qpv} f_v \tag{1}$$

In Eq. (1), w_{qpv} represents the contribution of the voxel v to the pth projection of the qth view. The definition of the contribution depends on the projection geometry. For laser absorption measurements a line integral geometry could be appropriate and w_{qpv} would be the intersection length of the line and the voxel. However, where each projection has an associated finite bin width (e.g., with a CCD) the fractional volume of the 3-D projection and the voxel is more appropriate for w_{qpv}, as illustrated in 2D in Fig. 1. The reconstruction quality

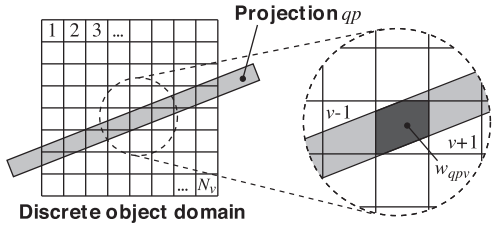

Fig. 1. A pixel's contribution w_{qpv} to projection qp is the intersection of the projection geometry and the pixel.

depends on the accurate approximation of the true volume integral through Eq. (1), and so a 3-D implementation of ART will improve accuracy. Determining the true projection volume can be challenging due to the camera optics and beam steering. In this work the projection geometry is approximated as either a cylinder or a square prism of a width that matches a view pixel p, but other work [39,40] has demonstrated how geometric optics can be used to include the effect of the camera optics and even refraction.

Equation (1) implies that the flame is optically thin and self-absorption of the emitted light is negligible, which is a reasonable assumption for visible emissions such as CH^* at 430 nm. For other emission wavelengths (e.g., the UV emissions of OH^*), self-absorption can be significant and must be considered in the reconstruction as demonstrated by Floyd and Kempf [24] or Correia et al. [15].

The iterative ART algorithm provides an optimum solution to the reconstruction problem by projecting back a diminishing error into the reconstruction domain until a convergence criterion is met. ART generates the error by comparing each measured projection value I_{qp} with an equivalent projection taken through the current iteration's (h) estimate of the object $\mathbf{f}^{(h)}$ using Eq. (1). Once all views have been addressed, the next iteration begins. Generally, a reasonable estimate of the object is produced after just one iteration, and higher levels of accuracy are obtainable with more iterations. (In this work 50–600 iterations were used.)

In this work a relaxed, additive version of ART is used as described in Eq. (2), where β is a relaxation factor (used to improve convergence in the presence of inevitable sensor noise), and where $\mathbf{w}_{qp}$ is a row vector consisting of w_{qpv} for all v.

$$\mathbf{f}^{(h+1)} = \mathbf{f}^{(h)} + \beta \mathbf{w}_{qp} \frac{I_{qp} - \mathbf{w}_{qp} \cdot \mathbf{f}^{(h)}}{\mathbf{w}_{qp} \cdot \mathbf{w}_{qp}} \tag{2}$$

Due to experimental errors, judging the convergence of ART is non-trivial, and the additive form of ART is selected over other forms for improved performance in this respect [37,41]. We employ the criterion of Mishra et al. [41], which considers a reconstruction converged once the absolute difference of the sum of the field f, from one iteration to the next, is below the threshold value $\Delta_c \times |\mathbf{f}^{(h)}|$ (typically Δ_c ranges from 10^{-3} to 10^{-6}). This simple criterion was chosen for its low computational cost, which is important for our large 3-D reconstructions.

The image reconstructions are based on noisy, incomplete data that includes some level of error, which is discussed elsewhere [39,42] in detail.

2.2. The Matrix burner

To develop and test the CTC sensor, a well defined non-axisymmetric test case was required that features multiple flame fronts. A steady Matrix burner was chosen, which consists of five incomplete rows of concentric fuel and oxidizer nozzles as shown in Fig. 2. The inner nozzles ($d_f = 1$ mm) supply 2 l/min of methane, while the outer nozzles ($d_o = 3$ mm) provide 35 l/min of oxygen. The entire burner provides up to 10 flame fronts in any planar direction, and so is representative of wrinkled turbulent flames.

3. High resolution measurements

This section reports a high resolution CTC reconstruction of the Matrix burner CH^* chemiluminescence performed using a single inexpensive camera, which was repositioned to provide the 48 views of the flame. It must be stressed that the Matrix burner flames are laminar and very steady, so that the reconstruction is independent of exposure time.

3.1. Specific experimental method

The single camera was mounted on a tripod and attached to an aluminium arm that was connected to the burner. The camera was rotated around the burner, while maintaining the alignment and a distance of 300 mm from the burner centre. Images were taken at equal angular intervals using a Minolta A1 2568 × 1928 pixel consumer CCD camera with a 14 bit A/D converter and a 200 mm macro lens at $f/11$ and a 200 ms exposure, which is sufficiently long to reduce imaging noise to

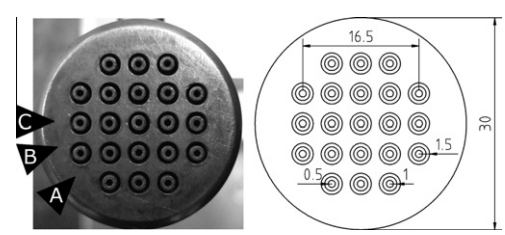

Fig. 2. A photograph of the Matrix burner showing the angles of views 12 (A), 19 (B) and 24 (C) seen in Fig. 3 (left) and a drawing of the Matrix burner (right).

a low level where it can no longer affect the reconstructions. To isolate the CH* emissions resulting from the $A^2\Delta - X^2\Pi$ transition, an interference band-pass filter of 430 ± 10 nm with 45% transmission at peak frequency was added in front of the lens. To avoid any artefacts from the camera's RGB Bayer filter, only the blue values from the un-interpolated raw images were used, yielding a view resolution of 48 μm.

For CT reconstruction, the precise centre of the burner in each image was required, and the images themselves were used for calibration. The left and right flame boundaries were determined using *edge detection*, with their mid-point taken as the burner centre. The rotation of the image about the camera's optical axis was removed by a counter rotation determined from the angle between the image horizontal and the bottom of the flame.

The CT algorithm requires that the w_{qpv} values are stored for each projection, and even exploiting the sparsity, more than 8 GB of memory would be required. To reduce the memory footprint, this high resolution reconstruction was performed by planes that were stacked afterwards. To speed up the reconstruction, the views were filtered from 450 pixels width to 225 pixels by binning two pixels in the horizontal direction. The reconstructions were performed on a grid of 230×230 pixels in 60 planes, resulting in voxel (volume elements) dimensions of 96 μm in the camera plane and 48 μm out of plane.

3.2. Results and discussion

Measurements of CH* chemiluminescent intensity were taken from 48 camera positions spaced equally over a total angle of 172° with an estimated positioning uncertainty of ±0.4°. A black screen was placed behind the burner to minimise the background signal. Example view measurements taken at three different angles can be seen in Fig. 3.

The reconstruction was performed for $\Delta_c = 1.0 \times 10^{-4}$ with square prism shaped projections. Sections of the reconstruction (Fig. 4) show that the constituent flames have been resolved and reconstructed at the correct position. The flame fronts show a correct shape in the longitudinal *yz* plane and are generally circular in the *xy* measurement plane, although some deformation is seen in the outer flames particularly in the top right of Fig. 4. This is thought to be due to errors in camera alignment, though it could also be a result of perspective distortion, limited depth-of-field, or refraction.

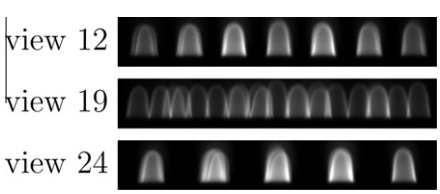

Fig. 3. Example views of the Matrix burner flame. Their viewing angles are depicted in Fig. 2.

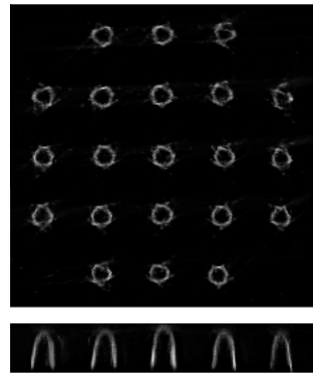

Fig. 4. *Top:* section through the reconstruction at 20 pixels (1 mm) above the burner exit plane. *Bottom:* section through the centre-line.

A volume rendering of the 3-D reconstruction is shown in Fig. 5 along with a photograph taken from the same angle. Floyd's analysis [39] estimates the wavelength resolved in the reconstruction to be 220 μm for 48 angles and a domain width of 22 mm. This seems a reasonable estimate given that the flame fronts have been resolved well. These results demonstrate that high resolution CTC can be performed effectively with inexpensive cameras.

4. Instantaneous measurements

This section reports on instantaneous measurements using ten simultaneous views from five inexpensive cameras and a suitable mirror arrangement.

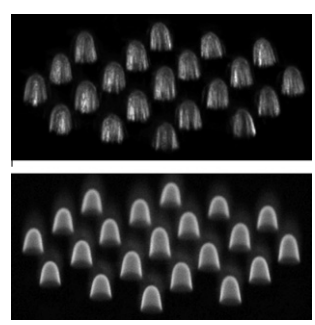

Fig. 5. A volume rendering of the 48 view reconstruction of the Matrix burner CH* intensity shown above a picture of the flame taken at a similar angle.

4.1. Specific experimental method

In this setup PicSight P32M cameras were used which feature a 656×494 pixel interline Sony ICX414 CCD, a 16 bit A/D converter, and a global shutter with a minimum exposure time of 10 μs. The cameras were fitted with 35 mm Pentax C-mount lenses, which have sufficient resolution for mega-pixel images, and were set to the maximum aperture of $f/1.6$ to maximise the signal. As the Matrix burner does not produce any soot, light is only emitted by chemiluminescence from the flame front, mainly by CH^*. It was therefore possible to perform the experiment without band-pass filters to maximise the amount of captured light and hence the signal-to-noise-ratio (SNR), with no negative effect on the reconstruction of the flame geometry. The cameras are small ($45 \times 32 \times 55$ mm) and cost approximately $800 each, making large deployments feasible.

Each camera was equipped with a system of mirrors that allowed it to simultaneously record two views 18° apart, giving a total of 10 views spanning 162°. The cameras and mirrors were all mounted to a single semi-annular 10 mm thick aluminium plate, as shown in Fig. 6, and each camera was connected to a common electronic external trigger to allow synchronous images to be taken. The mirrors were glass mounted, enhanced aluminium mirrors sized $80 \times 45 \times 3$ mm with a reported flatness of 2λ over 25 mm. The angles, locations and widths of the mirrors were determined using ray tracing to ensure that no vignetting was caused by the finite mirror width.

At an optical distance of 637 mm, the views sampled the 22 mm object domain approximately 118 times across its width, leading to a view resolution of approximately 190 μm per pixel and a depth-of-field of 22 mm. This depth-of-field is equal to the object depth and cylindrical projections are thus a good approximation to the true projection geometry [39]. For this instantaneous CTC setup the reconstruction was performed in full 3-D (without stacking) which allowed a correction for perspective distortion [39] to be applied. The correction is achieved by *fanning out* the projections, so they are no longer parallel, across an in-plane and out-of-plane angle determined by the optical distance and focal length of the lens.

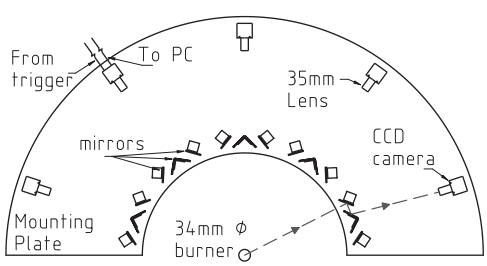

Fig. 6. The instantaneous CTC experimental setup. The five cameras are positioned 36° apart at a radius of 600 mm from the burner and connected to an external trigger. The mirrors provide two views for each camera, with an example ray path shown by the dashed line. All mirrors and cameras are mounted to a (10 mm) thick aluminium plate.

4.2. View registration

In general, the 3-D location and pose of each view needs to be known precisely. However, because of the common mounting plate, only the relative angle θ and the centre of each view must be determined in this case. To precisely determine θ and the view centre a cylindrical calibration object with an angular scale was used. The object has a diameter of 43.5 mm and was designed to *just fit* over the Matrix burner so that their central axes were aligned. The view centre was taken as the middle of the angular scale in the Y direction and the mid-point of the cylindrical calibration object in the X direction. The view angle θ was determined from the location of the view centre on the marked angular scale; to keep the precision high, sub-pixel edge detection was applied. The determined values varied from the designed, ideally spaced angles by at most 0.55°.

With any CT reconstruction, an error in the view location will increase the error in the reconstruction. To quantify the effect of error in θ, reconstructions have been performed using simulated views of a 2D *phantom* with random amounts of error in the viewing angles. The 232^2 pixel phantom consisted of a series of smoothed rings (Fig. 7), which represent a cross section of the Matrix burner flame. The error in the angle of each view was obtained using a random number generator that was scaled to the desired peak value ϵ_{peak}. For each peak value four reconstructions were performed to give a better idea of the spread of the error induced in the reconstruction. The correlation coefficient $r_{xy} = \mathrm{cov}(xy)/(\sigma_{xx}^2 \sigma_{yy}^2)$ of the phantom and reconstruction is used to quantify the error in the reconstruction.

The results of the test are shown in Fig. 8, where a negligible change in the correlation coef-

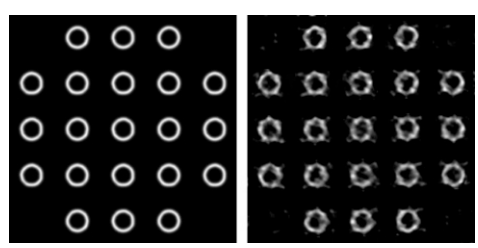

Fig. 7. The 2D phantom (left) and a 232^2 pixel reconstruction (right) with ten views and 2.3° of peak error (ϵ_{peak}) in θ.

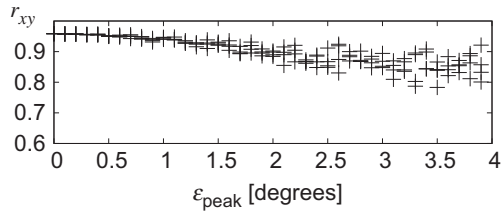

Fig. 8. The correlation coefficient r_{xy} of the phantom and reconstructions with increasing random error in θ.

ficient is seen for alignment errors of less than 0.5°, while acceptable correlation coefficients better than 0.9 are maintained up to 1.5° error. The reconstructions in Fig. 7 confirm this result, with significant deformation in the reconstruction only seen for alignment errors beyond 2.3°. The expected uncertainty of ±0.5° in the viewing angles will therefore not have a significant effect on the accuracy of the reconstructions.

4.3. The effect of noise

The iterative ART algorithms have been demonstrated in several previous studies [37,41,39] to converge to a solution in the presence of significant noise in the views, i.e., SNRs below 20 dB. Furthermore, the effect of noise on spatial resolution is small, as typically for CTC it is the number of views that has the most significant effect [43]. For CTC, the principle effect of noise is to increase the error in the reconstructions, because any noise/error in the projection measurements is passed to the CT algorithm and into the reconstruction. This relationship is illustrated by Fig. 9, which reports the error in reconstructions of the Matrix burner phantom (Fig. 7), obtained using 10 views with artificial noise added.

$$E_{RMS} = \frac{\sqrt{\sum(\mathbf{f} - \mathbf{f_p})^2 / N_v}}{\max(\mathbf{f_p})} \tag{3}$$

Figure 9 shows that a decrease in the view SNR leads to an increase in the reconstruction error E_{RMS} defined in Eq. (3), where $\mathbf{f_p}$ refers to the phantom object. The acceptable SNR of the views is thus set by the acceptable level of error in the reconstruction. Figure 9 shows that the effect of noise on the reconstruction error is small for SNRs above 20 dB.

4.4. Results and discussion

Simultaneous view measurements were performed using exposure times of 250 and 62 μs for the parameters shown in Table 1; even for the 62 μs exposure, a relatively high view SNR of 30 dB is achieved. Inspection of the views (e.g., Fig. 10) confirms the high SNRs, as the noise

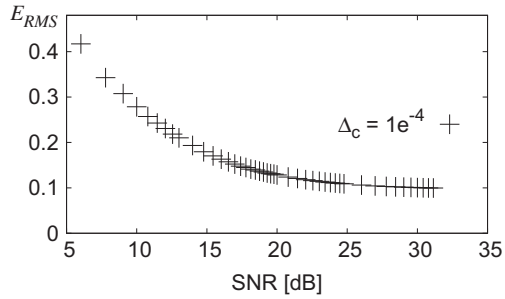

Fig. 9. The normalised RMS error, E_{RMS}, in reconstructions of the Matrix burner phantom obtained from 10

Table 1
Experimental parameters.

Exposure (μs)	Gain (dB)	SNR (dB)	Δ_c
250	15	40	1.0×10^{-4}
62	27	30	1.0×10^{-5}

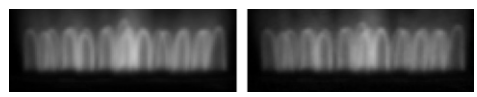

Fig. 10. View 7 for both exposures (left 250 μs, right 62 μs.)

level does not vary discernibly between the two exposure times. We have seen from Fig. 9 that an SNR of 20 dB produces only a moderate increase in error which suggests that even shorter exposures are possible. The views were cut to 115 × 42 pixels and reconstructed using the full 3-D ART algorithm with non-parallel, perspective corrected, cylindrical projections on a reconstruction grid of 115 × 115 × 42 voxels. Volume renderings and sections of the reconstructions are shown in Fig. 11.

Both reconstructions have well resolved the individual flames that are clearly distinct from each other. There is little difference between the 62 and 250 μs reconstructions due to the high SNR. The cavity within the flames is not as well resolved as with the high resolution reconstruction in Fig. 4, which is expected for the smaller number of views.

From the analysis of Floyd [39], the resolved wavelength for this 10 view setup should be approximately 2 mm, which agrees with the reconstructions which resolve the approximately 2 mm wide constituent flames. For turbulent flames of high Reynolds numbers, larger numbers of views must be used to ensure the resolution of the small scales. However, even under-resolved data will provide useful information, as demonstrated by Large–Eddy Simulation.

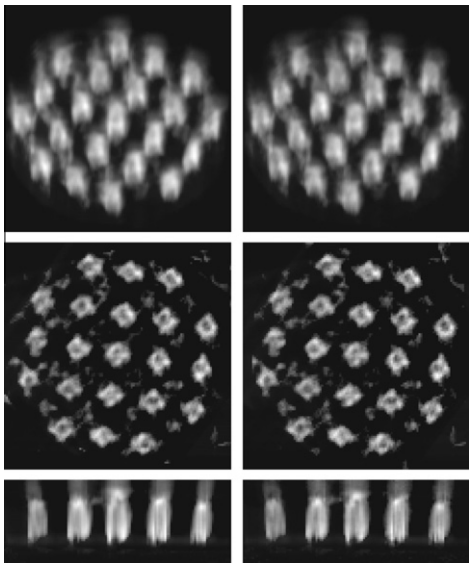

Fig. 11. *Top*: volume renderings of the 250 μs (left) and the 62 μs (right) reconstructions. *Middle*: *xy* section of each reconstruction at 25 pixels above the burner exit plane. *Bottom*: sections taken through the object centre.

Both instantaneous reconstructions are seen to suffer deformation similar to the *cross* shape observed in Fig. 7. This deformation is more severe in the reconstructions of the actual flame than in the phantom study, suggesting a greater error in θ. However, the difference may be partially explained by the larger diameter of the phantom's constituent flames. It is also possible that lens distortion contributes to the deformations, as in this setup the mirrors locate the flame towards the edges of the image (to avoid vignetting) where lens distortion effects are greatest. Such an error could be removed with additional calibration of the individual cameras as demonstrated by Willert [44]. If large numbers of cheap cameras are used, variations in the cameras' intensity response could also be determined and corrected to further improve the reconstructions.

The results presented here have demonstrated the ability of *affordable* cameras to provide short exposure, instantaneous views of sufficient SNR for ART reconstruction. While the use of pure oxygen and no optical filters has increased the signal strength, even shorter exposure times would have been possible with the removal of the mirrors to allow an increase in the sensor solid angle.

5. Conclusions

A technique that uses Computed Tomography of Chemiluminescence (CTC) to return instantaneous 3-D chemiluminescence intensities for arbitrary turbulent flames has been presented and tested for two setups. The first measures CH^* of a non-axisymmetric, steady Matrix burner flame to high resolution using 48 views from a single repositioned camera. The second measures the same burner but using 10 simultaneous views to return the instantaneous visible emissions profile for the flame.

Some localised error in the reconstructions is seen, mainly towards the reconstruction edges, and for high resolution applications more comprehensive view registrations may be necessary, along with more accurate projection geometries that account for the camera optics and refraction within the flame. The flexibility in the projection geometry that ART affords, however, means that such corrections have been demonstrated [39,40].

All reconstructions have shown good qualitative agreement with the observed shape of the flame. The high resolution reconstruction has resolved wavelengths of 220 μm and has shown the ability of CTC to resolve multiple flame fronts. Instantaneous reconstructions have been performed using inexpensive machine vision cameras for which large deployments (40+) are possible. The PicSight cameras have proved capable of returning simultaneous view measurements with a high enough SNR for reliable reconstruction at exposures as short as 62 μs, demonstrating the suitability of CTC for the measurement of turbulent flames.

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